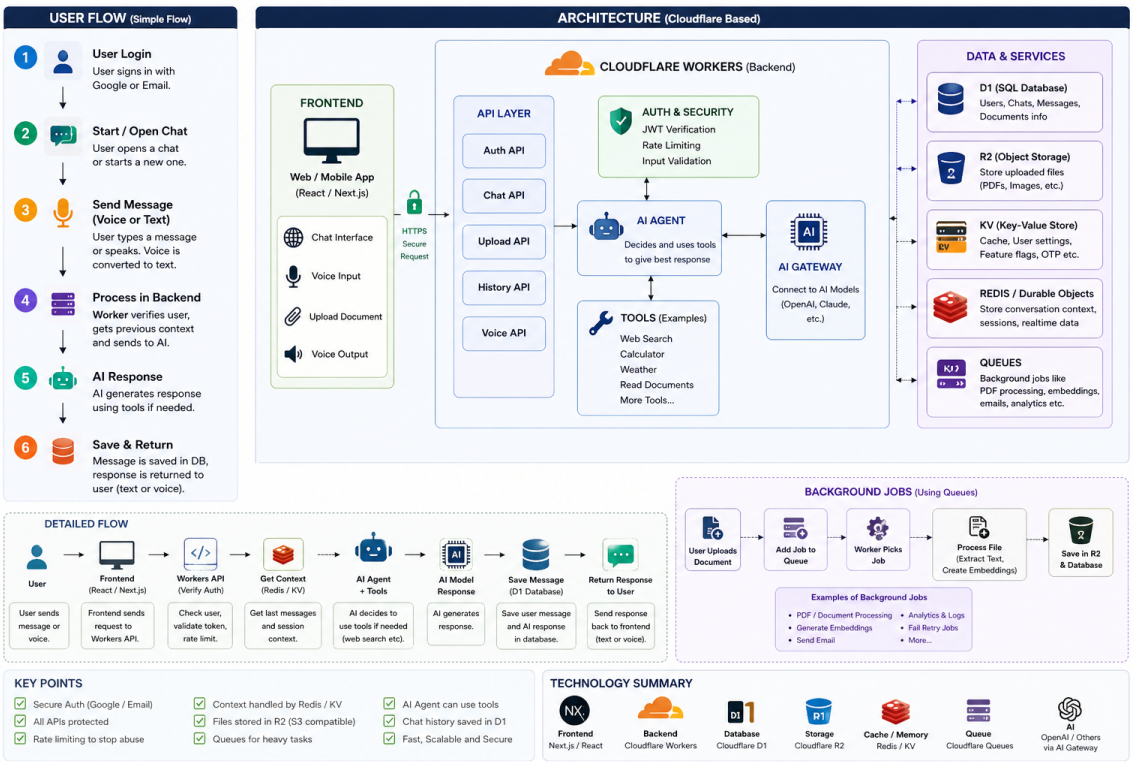
Flare GPT

A Cloudflare-First Full-Stack AI SaaS Platform

Author: Sahil Sarawgi `sahil.s23csai@nst.rishihood.edu.in` **Date:** April 29, 2026

Abstract. Flare GPT is a production-grade, serverless AI SaaS platform built entirely on Cloudflare's edge ecosystem. Users authenticate, upload documents, and chat with an AI agent that answers from their personal knowledge base — all without a single traditional server. The stack combines **Workers** (compute), **D1** (SQL), **R2** (object storage), **Vectorize** (vector search), **Workers AI** (inference), **Queues** (async jobs), **KV** (cache), and **AI Gateway** (model routing).

1. Architecture Diagram



[Insert Architecture Diagram Here]

Figure 1: Cloudflare-based architecture — User Flow · API Layer · AI Agent · Data Services · Background Jobs

2. System Overview & User Flow

Flare GPT follows a clean six-step request flow. **(1) Login:** users authenticate via Google OAuth or email/password; a Cloudflare Worker hashes credentials, writes to **D1**, and returns a signed **JWT**. **(2) Chat session:** a new session is opened in D1; recent context is loaded from **KV/Redis** for fast retrieval. **(3) Message:** text or voice is sent over HTTPS to the Chat Worker, where rate limiting is enforced. **(4) Processing:** the Worker verifies the JWT, embeds the query, and runs semantic search on **Vectorize** to retrieve relevant document chunks (RAG). **(5) AI Response:** the AI Agent — routed via **AI Gateway** — streams a grounded answer back to the user with optional tool calls (web search, calculator, weather). **(6) Persist:** the conversation is saved to D1 and returned as text or voice.

For document uploads, a fully async pipeline is triggered: the file is stored in **R2**, a job is pushed to **Cloudflare Queues**, text is extracted and chunked, then embedded via **Workers AI** and indexed in **Vectorize** — giving instant upload feedback while processing continues in the background.

3. Cloudflare Technologies

Service	Type	Role
Workers	Serverless Compute	All APIs (auth, chat, upload, history) via Hono framework
D1	SQL Database	Users, sessions, messages, document metadata
R2	Object Storage	Uploaded files (PDFs, images) — S3-compatible
Vectorize	Vector Index	Semantic search and RAG document retrieval
Workers AI	AI Inference	Text generation and embedding models at the edge
Queues	Message Queue	Async jobs: PDF parsing, embedding, emails, retries
KV	Key-Value Store	OTP codes, feature flags, session cache
AI Gateway	Model Proxy	Routes calls to OpenAI/Claude/Workers AI with logging

Conclusion

Flare GPT proves that a fully featured AI SaaS product — with secure auth, document intelligence, streaming responses, and global low-latency delivery — can be built using only Cloudflare's native services. No servers to manage, no manual scaling, and compute costs that stay proportional to real usage.